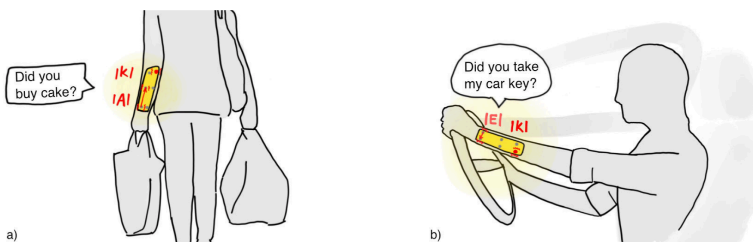
Tactual Communication - Skin Hearing

This is a continuation of my PhD work, under Building 8's flagship **Skin-Hearing** project. The two main questions were: 1. Is the skin capable of communicating high-bandwidth and high-throughput verbal information, and 2. Are tactual codes quickly learned, retained and scaled over time.

Coding Tactile Symbols for Phonemic Communication: We examine one's learning and processing capacity of broadband tactile information, such as that derived from speech. In Study 1, we tested a user's capability to recognize tactile locations and movements on the forearm in the presence of masking stimuli and determined 9 distinguishable tactile symbols. We associated these symbols to 9 phonemes using two approaches, random and articulation associations. Study 2 showed that novice participants can learn both associations. However, performance for retention, construction of words and knowledge transfer to recognize unlearned words was better with articulation association. In study 3, we trained novel participants to directly recognize words before learning phonemes. Our results show that novel users can retain and generalize the knowledge to recognize new words faster when they were directly trained on words. Finally, Study 4 examined optimal presentation rate for the tactile symbols without compromising learning and recognition rate.



	Stud 3 Day 1	Study 4
Word Recog	88.6 (11)	76.3 (14.6)
Word Recall	82.9 (23.6)	82.5 (8.7)
New Words	72.8 (25.6)	87.5 (12.6)
Phonemes	90.6 (7.7)	73.6 (11)

Table 4. Comparison between Study 3 Day 1 and Study 4 results. Word recognition accuracy was the average of all trials. Word recall and new words accuracy were from the last repetition. Standard deviations are in parentheses.

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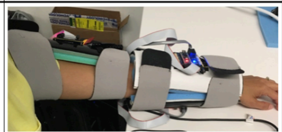
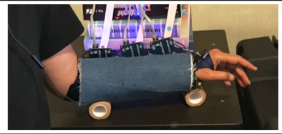
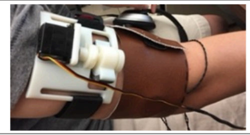
Through a series of partnerships with Purdue University, MIT and Rice University, we examined the skin's capacity to transfer verbal messages and the user's ability to learn and retain tactual codes while reducing the transfer time and contact area.

* Naive ** Experienced

Institute	Meta Platforms	Purdue University	Rice University
			
Body Site	dorsal upper arm + (dorsal + volar forearm)	dorsal + volar forearm	Upper arm (bands)
Actuators	24 VCM	24 VCM	4 VCM + Shear + Squeeze
Phonemes	13 (7C, 6V)	39 (24C, 15V)	23 (12C, 11V)
Number of Words	100 (1-3 phoneme words)	500 (2 single, 89 double, 359 triple, 49 quad, 1 pent)	50 (2 single, 14 double, 22 triple, 6 quad, 1 pent, 5 hex)
Training Period	65 min in 3 days	50-299 minutes** 266-490 minutes*	100 minutes in 4 days
Percentage Correct	86% (46%-100%) 6P > 90%	63-94%** 65%*	86.6%
Number of Participants	9*	11** 10*	10* (4M, 6F)
References	Chen 2018; Zhao 2018; Turcott 2018;	Tan 2020; Reed 2018	Sullivan 2018; Dunkelberger 2019

- Chen, J., Turcott, R., Castillo, P., Setiawan, W., Lau, F., & Israr, A. (2018, August). Learning to feel words: A comparison of learning approaches to acquire haptic words. In *Proceedings of the 15th ACM Symposium on Applied Perception* (pp. 1-7). [Link]
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- Reed, Charlotte M., et al. "A phonemic-based tactile display for speech communication." *IEEE transactions on haptics* 12.1 (2018): 2-17. [Link]
- Tan, Hong Z., et al. "Acquisition of 500 English words through a TActile phonemic sleeve (TAPS)." *IEEE Transactions on Haptics* 13.4 (2020): 745-760. [Link]
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- Sullivan, Jennifer L., et al. "Multi-sensory stimuli improve distinguishability of cutaneous haptic cues." *IEEE transactions on haptics* 13.2 (2019): 286-297. [Link]